

## MODULE - 4

—Shankargoud Patil

(1)

### Stability analysis and frequency domain analysis :

#### The concept of stability :

A linear time-invariant system is stable if the following two notions of system stability are satisfied.

- (i) When the system is excited by a bounded input, the output is bounded.
- (ii) In the absence of the input, the output tends towards zero (the equilibrium state of the system) irrespective of initial conditions. This stability concept is known as asymptotic stability.

Here, we are concerned with the stability determination of linear-time invariant (LTI) systems.

Let us observe the physical implication of the two notions of stability defined earlier, by considering a single-input, single-output system with transfer function

$$\frac{C(s)}{R(s)} = G(s) = \frac{b_0 s^m + b_1 s^{m-1} + \dots + b_m}{a_0 s^n + a_1 s^{n-1} + \dots + a_n} ; m < n$$

with initial conditions assumed zero, the output of the system is given by,

$$c(t) = \mathcal{L}^{-1} \{ G(s) R(s) \}$$

∴ Making use of convolution,

$$c(t) = \int_0^t g(\tau) x(t-\tau) d\tau$$

where  $g(t) = \mathcal{L}^{-1} \{ G(s) \}$  is the impulse response of the system.

Taking the absolute value on both sides we get,

$$|c(t)| = \left| \int_0^t g(\tau) x(t-\tau) d\tau \right|$$

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Since the absolute value of integral is not greater than the integral of the absolute value of the integrand,

$$|c(t)| \leq \int_0^a |\xi(\tau) x(t-\tau)| d\tau$$

$$\leq a \int_0^a |\xi(\tau)| \cdot |x(t-\tau)| d\tau$$

The first notion of stability is satisfied if for every bounded input ( $|x(t)| \leq M_1 < \infty$ ), the output is bounded ( $|c(t)| \leq M_2 < \infty$ ).

From the above eq<sup>n</sup>, we have for bounded input, the bounded output condition as,

$$|c(t)| \leq M_1 \int_0^a |\xi(\tau)| d\tau \leq M_2$$

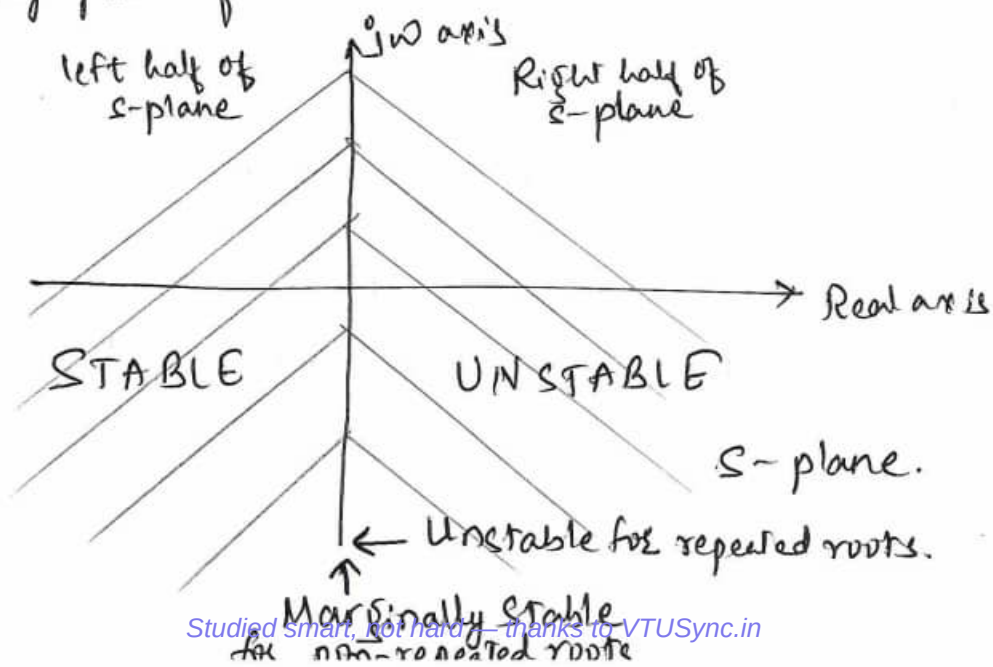
Thus the first notion of stability is satisfied if the impulse response  $\xi(t)$  is absolutely integrable.

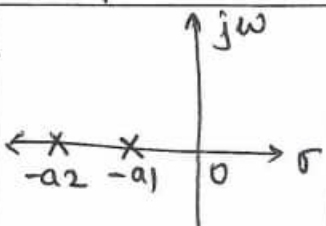
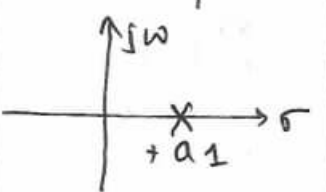
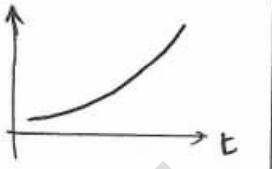
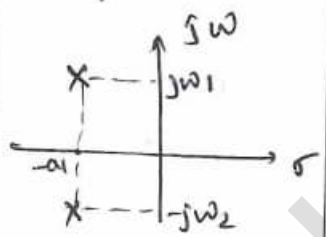
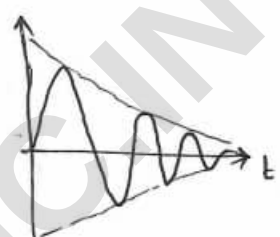
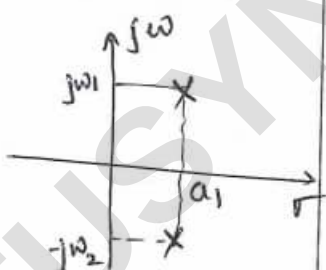
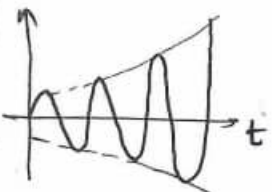
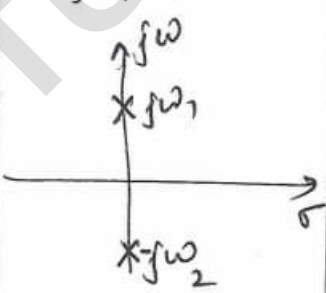
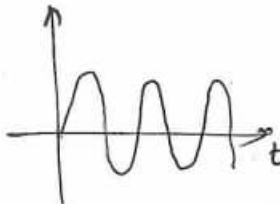
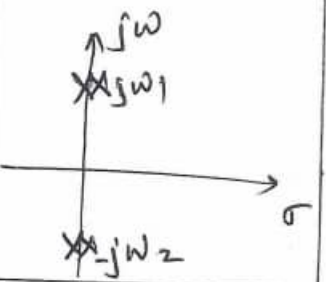
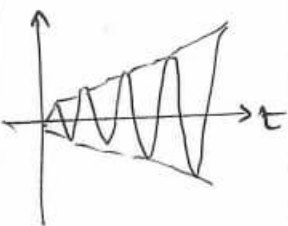
$$\text{i.e. } \int_0^a |\xi(\tau)| d\tau < \infty$$

Stability depends on locations of closed-loop poles. The closed-loop poles are the roots of the characteristic equation of the system.

$\therefore$  Closed-loop poles = Roots of the characteristic eq<sup>n</sup>

S-plane can be divided into three distinct zones from stability point of view as shown below.



| Sl. No. | Nature of closed loop poles                                            | Location of closed loop poles in s-plane                                            | Step Response                                                                        | Stability condition             |
|---------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------|
| 1.      | Real and Negative<br>i.e. in LHS of s-plane                            |    |    | Absolutely stable               |
| 2.      | Real and Positive<br>i.e. in RHS of s-plane                            |    |    | Unstable                        |
| 3.      | Complex conjugate with negative real part i.e. in LHS of s-plane       |    |    | Absolutely stable               |
| 4.      | Complex conjugate with Positive real part i.e. in RHS of s-plane       |   |  | Unstable                        |
| 5.      | Non repeated pair on imaginary axis without any pole in RHS of s-plane |  |  | Marginally or Critically stable |
| 6.      | Repeated pair on Imaginary axis without any pole in RHS of s-plane     |  |  | Unstable.                       |

Closed loop poles and stability



## Necessary conditions for stability

Certain conclusions regarding the stability of a system can be drawn by merely inspecting the coefficients of its characteristic eq<sup>n</sup> in polynomial form. In the following sections, we shall show that a necessary (but not sufficient) condition for stability of a linear system is that all coefficients of its characteristic equations  $q(s)=0$ , be real and have the same sign. Furthermore, none of the coefficients should be zero.

In simple words,

- (1) All the coefficients of the polynomial have the same sign.
  - (2) None of the coefficient vanishes i.e all powers of 's' must be present in descending order from 'n' to zero.
- These These conditions are necessary but not sufficient.

## HURWITZ stability criterion :

The characteristic eq<sup>n</sup> of an  $n$ th order system is,

$$q(s) = a_0 s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n = 0$$

For the stability of this system, it is necessary and sufficient that the 'n' determinants formed from the coefficients  $a_0, a_1, \dots, a_n$  of the characteristic eq<sup>n</sup> be positive, where these determinants are taken as the principal minors of the following arrangement (called the Hurwitz determinant):

$$H = \begin{vmatrix} a_1 & a_0 & 0 & 0 & 0 & 0 \dots 0 & 0 \\ a_3 & a_2 & a_1 & a_0 & 0 & 0 \dots 0 & 0 \\ a_5 & a_4 & a_3 & a_2 & a_1 & a_0 \dots 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{2n-1} & a_{2n-2} & a_{2n-3} & \dots & \dots & a_{n-1} & a_n \end{vmatrix}$$

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where the coefficients with indices larger than 'n' or with negative indices are replaced by zeros.

In other words, the necessary and sufficient conditions for stability are,

$$\Delta_1 = a_1 > 0;$$

$$\Delta_2 = \begin{vmatrix} a_1 & a_0 \\ a_3 & a_2 \end{vmatrix} > 0;$$

$$\Delta_3 = \begin{vmatrix} a_1 & a_2 & 0 \\ a_3 & a_2 & a_1 \\ a_5 & a_4 & a_3 \end{vmatrix} > 0; \dots$$

$$\Delta_n = \text{entire arrangement of } H > 0$$

Numericals:

- ① Consider a fourth order system with the characteristic eq<sup>n</sup>  
 $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$

Sol<sup>n</sup>:

$$a_0 = 1, a_1 = 8, a_2 = 18, a_3 = 16, a_4 = 5$$

The Hurwitz arrangement is given below:

$$\begin{array}{c|cccc} a_1 & 8 & 1 & 0 & 0 \\ a_3 & 16 & 18 & 8 & 1 \\ a_5 & 0 & 5 & 16 & 18 \\ a_7 & 0 & 0 & 0 & 5 \end{array}$$

Therefore,  $\Delta_1 = a_1 = 8 > 0;$

$$\Delta_2 = \begin{vmatrix} 8 & 1 \\ 16 & 18 \end{vmatrix} = (8 \times 18) - (16 \times 1) = 128 > 0$$

$$\Delta_3 = \begin{vmatrix} 8 & 1 & 0 \\ 16 & 18 & 8 \\ 0 & 5 & 16 \end{vmatrix} = [8(18 \times 16) - (8 \times 5)] - i[(16 \times 16) - 0] + 0$$

$$= 1984 - 256$$

$$\Delta_3 = 1728 > 0$$

$$\Delta_4 = \begin{vmatrix} 8 & 1 & 0 & 0 \\ 16 & 18 & 8 & 1 \\ 0 & 5 & 16 & 18 \\ 0 & 0 & 0 & 5 \end{vmatrix}$$

$$\begin{aligned}
 &= 8 \begin{vmatrix} 18 & 8 & 1 \\ 5 & 16 & 18 \\ 0 & 0 & 5 \end{vmatrix} - 1 \begin{vmatrix} 16 & 8 & 1 \\ 0 & 16 & 18 \\ 0 & 0 & 5 \end{vmatrix} + 0 - 0 \\
 &= 8 \left[ \{18(16 \times 5) - 0\} - 8\{5 \times 5 - 0\} + 1\{0 - 0\} \right] - \\
 &\quad 1 \left[ \{16(16 \times 5 - 0)\} - 8\{0 - 0\} + 1\{0\} \right] \\
 &= 8[1440 - 200] - 1[1280 - 0 + 0] \\
 &= 8640 > 0 ;
 \end{aligned}$$

Therefore the system under consideration is stable.

② Determine the stability of the given characteristic eq<sup>n</sup> by Hurwitz's method.

$F(s) = s^3 + s^2 + s + 4 = 0$  is a characteristic equation.

Sol<sup>n</sup>:

$$a_0 = 1, a_1 = 1, a_2 = 1, a_3 = 4$$

The Hurwitz arrangement is given below.

$$H = \begin{vmatrix} a_1 & a_3 & 0 \\ a_0 & a_2 & a_4 \\ 0 & 0 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 0 \\ 4 & 1 & 1 \\ 0 & 0 & 4 \end{vmatrix}$$

$$\text{Therefore, } \Delta_1 = 1 > 0 ;$$

$$\Delta_2 = \begin{vmatrix} 1 & 1 \\ 4 & 1 \end{vmatrix} = 1 - 4 = -3 < 0$$

$$\Delta_3 = \begin{vmatrix} 1 & 1 & 0 \\ 4 & 1 & 1 \\ 0 & 0 & 4 \end{vmatrix}$$

$$= 1(4) - 1(16) + 0$$

$$\Delta_3 = -12 < 0$$

Since  $\Delta_2$  and  $\Delta_3$  are negative, the system under consideration is Unstable.



## Disadvantages of Hurwitz's method

- (i) For higher order systems, to solve the determinants of higher order is very complicated and time consuming.
- (ii) Number of roots located in right half of s-plane for unstable system cannot be judged by this method.
- (iii) Difficult to predict marginal stability of the system.

Due to these limitations, a new method is suggested by the scientist Routh called Routh's method. It is also called Routh-Hurwitz's method.

## Routh Stability Criterion:

This criterion is based on ordering the coefficients of the characteristic equation into an array, called the Routh array as given below:

$$q(s) = a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_{n-1} s + a_n = 0$$

### Routh array:

|           |       |       |       |       |   |   |
|-----------|-------|-------|-------|-------|---|---|
| $s^n$     | $a_0$ | $a_2$ | $a_4$ | $a_6$ | . | . |
| $s^{n-1}$ | $a_1$ | $a_3$ | $a_5$ | .     | . | . |
| $s^{n-2}$ | $b_1$ | $b_2$ | $b_3$ | .     | . | . |
| $s^{n-3}$ | $c_1$ | $c_2$ | $c_3$ | .     | . | . |
| $s^{n-4}$ | $d_1$ | $d_2$ |       |       |   |   |
| .         | .     | .     |       |       |   |   |
| .         | .     | .     |       |       |   |   |
| $s^2$     | $e_1$ | $a_n$ |       |       |   |   |
| $s^1$     | $f_1$ |       |       |       |   |   |
| $s^0$     | $a_n$ |       |       |       |   |   |

The coefficients  $b_1, b_2, \dots$  are evaluated as follows:

$$b_1 = (a_1 a_2 - a_0 a_3) / a_1 ;$$

$$b_2 = (a_1 a_4 - a_0 a_5) / a_1 ; \dots$$

This process is continued till we get a zero as the last coefficient in the third row. In a similar way, the coefficients of  $4^{th}$ ,  $5^{th}$ , ...,  $n^{th}$  and  $(n+1)^{th}$  rows are evaluated, e.g.,

$$c_1 = (b_1 a_3 - a_1 b_2) / b_1;$$

$$c_2 = (b_1 a_5 - a_1 b_3) / b_1; \dots$$

and

$$d_1 = (c_1 b_2 - b_1 c_2) / c_1;$$

$$d_2 = (c_1 b_3 - b_1 c_3) / c_1; \dots$$

It is to be noted here that in the process of generating the Routh array, the missing terms are regarded as zero.

Also all the elements of any row can be divided by a positive constant during the process to simplify the computational work.

The Routh stability criterion is stated as below.

For a system to be stable, it is necessary and sufficient that each term of first column of Routh array of its characteristic equation be positive if  $a_0 > 0$ . If this condition is not met, the system is unstable and number of sign changes of the terms of the first column of the Routh array corresponds to the number of roots of the characteristic equation in the right half of the  $s$ -plane.



Numericals:

- ① Examine the stability of  $F(s) = s^3 + 6s^2 + 11s + 6 = 0$  using Routh stability criterion.

Sol<sup>n</sup>: The Routh array for the system is as given below.

$$\begin{array}{c|cc}
 s^3 & 1 & 11 \\
 s^2 & 6 & 6 \\
 s^1 & \frac{11 \times 6 - 6 \times 1}{6} = 10 & 0 \\
 s^0 & \frac{10 \times 6 - 6 \times 0}{10} = 6 & 
 \end{array}$$

As there is no sign change in first column, system is stable.

- ② Consider the fourth-order system with the characteristic eq<sup>n</sup>  $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$ . Comment on the stability using Routh criterion.

Sol<sup>n</sup>:  $a_0 = 1, a_1 = 8, a_2 = 18, a_3 = 16, a_4 = 5$

The Routh array is,

$$\begin{array}{c|ccc}
 s^4 & 1 & 18 & 5 \\
 s^3 & 8 & 16 & 0 \text{ (for missing term)} \\
 s^2 & \frac{8 \times 18 - 16 \times 1}{8} = 16 & \frac{8 \times 5 - 1 \times 0}{8} = 5 & \\
 s^1 & \frac{16 \times 16 - 8 \times 5}{16} = 135 & 0 & \\
 s^0 & \frac{135 \times 5 - 16 \times 0}{135} = 5 & & 
 \end{array}$$

The elements of the first column are all positive and hence the system is stable.

- ③ Comment on the stability of the system whose characteristic eq<sup>n</sup> is,  $3s^4 + 10s^3 + 5s^2 + 5s + 2 = 0$ , using Routh's criterion.

Sol<sup>n</sup>:

$$3s^4 + 10s^3 + 5s^2 + 5s + 2 = 0$$

$$a_0 = 3, a_1 = 10, a_2 = 5, a_3 = 5, a_4 = 2$$

|       |                                                      |                                           |   |
|-------|------------------------------------------------------|-------------------------------------------|---|
| $s^4$ | 3                                                    | 5                                         | 2 |
| $s^3$ | 10                                                   | 5                                         | 0 |
| $s^2$ | $\frac{10 \times 5 - 3 \times 5}{10} = 3.5$          | $\frac{10 \times 2 - 3 \times 0}{10} = 2$ |   |
| $s^1$ | $\frac{3.5 \times 5 - 10 \times 2}{3.5} = -0.71$     | 0                                         |   |
| $s^0$ | $\frac{-0.71 \times 2 - 3.5 \times 0}{-0.71} = 6.09$ |                                           |   |

Examining the first column of the Routh array, it is found that there are two changes in the sign. Therefore the system under consideration is unstable having Two poles in the right half of the s-plane.

## Special cases:

Occasionally, in applying the Routh stability criterion, certain difficulties arise causing the breakdown of the Routh's test. The difficulties encountered are generally of the following types.

Difficulty 1: When the first term in any row of the Routh array is zero while rest of the row has at least one non-zero term.

Because of this zero term, the terms in the next row become infinite and Routh's test breaks down. The following methods can be used to overcome this difficulty.

- Substitute a small positive number  $\epsilon$  for the zero and proceed to evaluate the rest of the Routh array. Then examine the signs of the first column of Routh array by letting  $\epsilon \rightarrow 0$ .
- Modify the original characteristic equation by replacing  $s$  by  $1/z$ . Apply the Routh's test on the modified equation in terms of  $z$ . This method works in most but not all cases.

## Numericals:

- Comment on stability by considering the characteristic eq<sup>n</sup>  $s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$  using Routh's criterion.

Soln: The Routh array is,  $a_0=1, a_1=1, a_2=2, a_3=2, a_4=3, a_5=5$

|       |                                                                    |                                          |   |
|-------|--------------------------------------------------------------------|------------------------------------------|---|
| $s^5$ | 1                                                                  | 2                                        | 3 |
| $s^4$ | 1                                                                  | 2                                        | 5 |
| $s^3$ | $\frac{1 \times 2 - 1 \times 2}{1} = 0$<br>$\epsilon$              | $\frac{1 \times 3 - 1 \times 5}{1} = -2$ |   |
| $s^2$ | $\frac{2\epsilon + 2}{\epsilon}$                                   | $\frac{5\epsilon - 0}{\epsilon} = 5$     |   |
| $s^1$ | $\frac{4\epsilon - 4 - 5\epsilon^2}{2\epsilon + 2} \rightarrow -2$ |                                          |   |
| $s^0$ | 5                                                                  |                                          |   |



Examining the terms in the first column of the Routh array, it is found that there are two changes in sign and hence the system is Unstable having two poles in the right half of s-plane.

The same numerical can also be solved by the second method by replacing  $s$  by  $1/z$ .

The characteristic eq<sup>n</sup> will become like,

$$5\left(\frac{1}{z}\right)^5 + \left(\frac{1}{z}\right)^5 + \left(\frac{1}{z}\right)^4 + 2\left(\frac{1}{z}\right)^3 + 2\left(\frac{1}{z}\right)^2 + 3\left(\frac{1}{z}\right) + 5 = 0$$

$$1 + z + 2z^2 + 2z^3 + 3z^4 + 5z^5 = 0$$

$$\therefore 5z^5 + 3z^4 + 2z^3 + 2z^2 + z + 1 = 0$$

The Routh array for this eq<sup>n</sup> will be

|       |        |        |   |
|-------|--------|--------|---|
| $z^5$ | 5      | 2      | 1 |
| $z^4$ | 3      | 2      | 1 |
| $z^3$ | $-4/3$ | $-2/3$ |   |
| $z^2$ | $1/2$  | 1      |   |
| $z^1$ | 2      |        |   |
| $z^0$ | 1      |        |   |

The system is Unstable with two poles in right half of s-plane.

### Difficulty 2:

When all the elements in any one row of the Routh array are zero. Here the Routh's test breaks down. This situation is overcome by replacing the rows of zeros in the Routh array by a row of coefficients of the polynomial generated by taking the first derivative of the auxiliary polynomial.

Auxiliary Polynomial: The polynomial whose coefficients are the elements of the row just above the

row of zeros in the Routh array is called an auxiliary polynomial.

Numericals:

- ① Consider a sixth-order system whose characteristic eq<sup>n</sup> is,  $s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$

Sol<sup>n</sup>:

$$a_0 = 1, a_1 = 2, a_2 = 8, a_3 = 12, a_4 = 20, a_5 = 16, a_6 = 16$$

The Routh array is,

|       |   |    |    |    |
|-------|---|----|----|----|
| $s^6$ | 1 | 8  | 20 | 16 |
| $s^5$ | 2 | 12 | 16 |    |
| $s^4$ | 2 | 12 | 16 |    |
| $s^3$ | 0 | 0  |    |    |
| $s^2$ |   |    |    |    |
| $s^1$ |   |    |    |    |
| $s^0$ |   |    |    |    |

Since the terms in the  $s^3$ -row are all zero, the Routh's test breaks down. Now the auxiliary polynomial is formed from the coefficients of the  $s^4$ -row, which is given by,

$$A(s) = 2s^4 + 12s^2 + 16$$

The derivative of the polynomial with respect to  $s$  is

$$8s^3 + 24s + 0 = 8s^3 + 24s$$

The zeros in the  $s^3$ -row are now replaced by the coefficients 8 and 24. The Routh array then becomes,

|       |      |    |    |    |
|-------|------|----|----|----|
| $s^6$ | 1    | 8  | 20 | 16 |
| $s^5$ | 2    | 12 | 16 |    |
| $s^4$ | 2    | 12 | 16 |    |
| $s^3$ | 8    | 24 |    |    |
| $s^2$ | 6    | 16 |    |    |
| $s^1$ | 2.66 |    |    |    |
| $s^0$ | 16   |    |    |    |

As there is no change of sign, the system may be stable. But there is row of zero, the system can be (i) Marginally stable or (ii) Unstable.

To examine this solve  $A(s) = 0$

$$2s^4 + 12s^2 + 16 = 0$$

$$s^4 + 6s^2 + 8 = 0$$

Put  $s^2 = y$

$$\therefore y^2 + 6y + 8 = 0$$

$$y = \frac{-6 \pm \sqrt{36 - 32}}{2}$$

$$= -3 \pm 1$$

$$= -2, -4$$

$$s^2 = -2 \text{ \& } s^2 = -4$$

$$\therefore s = \pm j\sqrt{2} \text{ \& } s = \pm j2$$

Non repeated roots on imaginary axis. Hence system is marginally stable.



- ② Check the stability of the given characteristic equation using Routh's method and also determine how many roots lie in 1) RHS 2) LHS and 3) On  $j\omega$  axis.  $P(s) = s^6 + s^5 + 5s^4 + s^3 + 2s^2 - 2s - 8$

Sol<sup>n</sup>:

$$a_0 = 1, a_1 = 1, a_2 = 5, a_3 = 1, a_4 = 2, a_5 = -2, a_6 = -8$$

The Routh's array is,

|       |   |   |    |    |
|-------|---|---|----|----|
| $s^6$ | 1 | 5 | 2  | -8 |
| $s^5$ | 1 | 1 | -2 | 0  |
| $s^4$ | 4 | 4 | -8 | 0  |
| $s^3$ | 0 | 0 | 0  |    |
| $s^2$ |   |   |    |    |
| $s^1$ |   |   |    |    |
| $s^0$ |   |   |    |    |

The Routh's test breaks down here. Now, the auxiliary polynomial is formed from the coefficients of the  $s^4$ -row, which is given by,

$$A(s) = 4s^4 + 4s^2 - 8$$

The derivative of the polynomial w.r.t 's' is,

$$16s^3 + 8s$$

The zeros in the  $s^3$ -row are now replaced with the coefficients of 16 & 8.

|       |    |    |    |    |
|-------|----|----|----|----|
| $s^6$ | 1  | 5  | 2  | -8 |
| $s^5$ | 1  | 1  | -2 | 0  |
| $s^4$ | 4  | 4  | -8 | 0  |
| $s^3$ | 16 | 8  | 0  |    |
| $s^2$ | 2  | -8 | 0  |    |
| $s^1$ | 72 | 0  |    |    |
| $s^0$ | -8 | 0  |    |    |

The system is Unstable as there is one sign change.

Solving  $p(s) = 0$

$$4s^4 + 4s^2 - 8 = 0$$

$$s^4 + s^2 - 2 = 0$$

Put  $s^2 = y$

$$y^2 + y - 2 = 0$$

$$y = \frac{-1 \pm \sqrt{1+8}}{2}$$

$$= \frac{-1 \pm 3}{2}$$

$$= -2, 1$$

$$\Rightarrow s^2 = -2 \text{ and } s^2 = 1$$

$$\therefore s = \pm j\sqrt{2}, s = \pm 1$$

1) RHS  $\rightarrow$  One Root.

2) LHS  $\rightarrow$  3 Roots

3) jw axis  $\rightarrow$  2 Roots